

Math727 – Numerical Analysis II

Homework 2

1. (a) Localize the spectrum of the matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 7 & -1 \\ -1 & 0 & 5 \end{bmatrix}$$

- (b) Provide an estimate of the number of complex eigenvalues of the matrix

$$\mathbf{A} = \begin{bmatrix} -4 & 0 & 0 & 0.5 & 0 \\ 2 & 2 & 4 & -3 & 1 \\ 0.5 & 0 & -1 & 0 & 0 \\ 0.5 & 0 & 0.2 & 3 & 0 \\ 2 & 0.5 & -1 & 3 & 4 \end{bmatrix}$$

2. (a) Let $\mathbf{A} \in \mathbb{C}^{N \times N}$ be a diagonal matrix and let $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{E}$ be a perturbation of $\mathbf{A}$ with $e_{ii} = 0$ for $i = 1, \dots, N$. Show that

$$|\lambda_i(\tilde{\mathbf{A}}) - \lambda_i(\mathbf{A})| \leq \sum_{j=1}^N |e_{ij}|, \quad i = 1, \dots, N,$$

when $\|\mathbf{E}\|_2$ is sufficiently small.

- (b) Verify (a) to the case in which

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \quad \mathbf{E} = \begin{bmatrix} 0 & \epsilon \\ \epsilon & 0 \end{bmatrix}$$

for some $\epsilon \geq 0$.

3. (a) Suppose that all the assumptions needed to apply the power method are satisfied except for the requirement $\alpha_1 \neq 0$. Show that in such a case the sequences generated by

$$\begin{aligned} \mathbf{z}^{(k)} &= \mathbf{A} \mathbf{q}^{(k-1)} \\ \mathbf{q}^{(k)} &= \mathbf{z}^{(k)} / \|\mathbf{z}^{(k)}\|_2 \\ \mu^{(k)} &= \mathbf{q}^{(k)H} \mathbf{A} \mathbf{q}^{(k)} \end{aligned}$$

converges to the eigenvalue/eigenvector pair $(\lambda_2, \mathbf{x}_2)$.

- (b) (Programming exercise) Program the power method described in Algorithm 8.7 of lecture notes. Compute the pair $(\lambda_1, \mathbf{x}_1)$ for the matrix

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 2 \\ -2 & 0 & 5 \\ 6 & -3 & 6 \end{bmatrix}$$

Submit the codes and print out the results with taking $\mathbf{q}^{(0)} = [1, 1, 1]^T / \sqrt{3}$ and $\mathbf{q}^{(0)} = \mathbf{w}^{(0)} / \|\mathbf{w}^{(0)}\|_2$ where $\mathbf{w}^{(0)} = [-3, -\frac{26}{3}, -\frac{32}{3}]^T$, respectively.

4. For a symmetric matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$ and a vector $\mathbf{x} = [x_1, x_2, \dots, x_N]^T \in \mathbb{R}^N$ with $x_j \neq 0$ for $j = 1, \dots, N$, let

$$d_j = \frac{(\mathbf{A}\mathbf{x})_j}{x_j}, \quad j = 1, \dots, N.$$

Show that the interval $[\mu - \rho, \mu + \rho]$ with $\rho = \max_{1 \leq j \leq N} |d_j - \mu|$ contains at least one eigenvalue λ for every number $\mu \in \mathbb{R}$.

5. Let $\mathbf{A}, \Delta\mathbf{A} \in \mathbb{R}^{N \times N}$ be symmetric matrices. If $\Delta\mathbf{A}$ is positive definite, show that

$$\lambda_j(\mathbf{A}) \leq \lambda_j(\mathbf{A} + \Delta\mathbf{A}), \quad j = 1, \dots, N$$

where

$$\begin{aligned} \lambda_1(\mathbf{A}) &\geq \lambda_2(\mathbf{A}) \geq \dots \geq \lambda_N(\mathbf{A}), \\ \lambda_1(\mathbf{A} + \Delta\mathbf{A}) &\geq \lambda_2(\mathbf{A} + \Delta\mathbf{A}) \geq \dots \geq \lambda_N(\mathbf{A} + \Delta\mathbf{A}). \end{aligned}$$

6. (Programming exercise) For the matrix $\mathbf{A} = (a_{ij}) \in \mathbb{R}^{N \times N}$ with

$$a_{ij} = \begin{cases} N - i + 1, & \text{if } j \leq i, \\ N - j + 1, & \text{otherwise,} \end{cases}$$

and using the LR algorithm, numerically determine for $N = 50$ and $N = 100$ its respective smallest and largest (by module) eigenvalues. Let $\mathbf{A}^{(k)} = (a_{ij}^{(k)})$, stop the method if $k \geq 100$ or

$$\epsilon_k = \max_{1 \leq j \leq N} \frac{|a_{jj}^{(k-1)} - a_{jj}^{(k)}|}{|a_{jj}^{(k-1)}|} \leq 0.05$$

is satisfied. Submit your codes. Provide the obtained approximations for the eigenvalues and also the values $\{\epsilon_k\}$ in a table.

Appendix: (LR Algorithm)

$\mathbf{A}^{(0)} = \mathbf{A}$

for $k = 1, 2, \dots$

Determine the factorization $\mathbf{A}^{(k-1)} = \mathbf{L}^{(k)}\mathbf{R}^{(k)}$ where

$\mathbf{L}^{(k)}$ and $\mathbf{R}^{(k)}$ are lower and upper triangular matrices respectively

Set $\mathbf{A}^{(k)} = \mathbf{R}^{(k)}\mathbf{L}^{(k)}$

end

Remark: The convergence analysis for the LR method is similar to the QR method.